

Problem 10.38

Show that

$$\frac{d}{dt}\delta_t = \frac{A''(t)}{A(t)} - \delta_t^2.$$

Solution

The force of interest at time t is

$$\delta_t = \frac{A'(t)}{A(t)}.$$

Differentiate both sides with respect to t . By the quotient rule,

$$\frac{d}{dt}\delta_t = \frac{A''(t)A(t) - A'(t)A'(t)}{[A(t)]^2}.$$

Separate the two terms:

$$\frac{d}{dt}\delta_t = \frac{A''(t)}{A(t)} - \frac{[A'(t)]^2}{[A(t)]^2}.$$

Since

$$\frac{A'(t)}{A(t)} = \delta_t,$$

we have

$$\frac{[A'(t)]^2}{[A(t)]^2} = \delta_t^2.$$

Therefore,

$$\boxed{\frac{d}{dt}\delta_t = \frac{A''(t)}{A(t)} - \delta_t^2}.$$